



Roll No

NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

Mid Semester Examinations, September-2025

Course Name: **Engineering Drawing**

Date: 14-10-2025

Duration: 100 Minutes

Course Code: **ME101**

Time: 11:15 AM to 12:55 PM

Max. Marks: 50

Instructions: Answer all the questions. If additional data is required, assume it suitably and indicate the assumption in your solution. Use only instruments approved for technical drawing: compass, divider, set squares, protractor, T-square, French curves, etc. All construction lines must be light and thin, and final lines must be dark and bold. Label all views, points, and dimensions clearly and neatly. Do not use calculators or electronic devices.

1.

- (a) State whether the statement is true or false, and justify your answer. (2)
- (i) If the projections of a line on both the horizontal and vertical planes are known, can you always determine the true length of the line?
- (ii) If the VT lies below the XY line, it indicates that the line intersects the vertical plane in the first quadrant.
- (b) What would be the position of line AB, if its projection results in all its key points (VT, HT, a' , b' , a , b , v , t) lying on a single straight line in the orthographic drawing. (HT: horizontal trace, VT: vertical trace, a' : front view projection of end point A, b' : front view projection of end point B, a : top view projection of end point A, b : top view projection of end point B, v : projector of VT, h : projector of HT) (2)
- (c) What are the fundamental differences between first-angle projection and third-angle projection. (2)
- (d) A point P is situated in the first quadrant. It is 30 mm above HP and 40 mm in front of VP. Draw its orthographic projections. (4)

2. A 75 mm long straight line lies in the first quadrant. It is inclined at 60° to the VP, lies in the HP and one end is in the VP. Draw its orthographic projections. (10)
3. The end A of a line AB is 10 mm in front of VP and 20 mm above HP. The line is inclined at 30° to HP and front view is 45° with XY. Top view is 60 mm long. Draw the projections. Find the true length and inclination with VP. Locate the traces. (10)
4. Determine the true length and true inclinations (θ and ϕ) of line PQ using the auxiliary planes method, based on the provided data. The horizontal trace (HT) of the line is 40 mm below the XY line (in the HP). The vertical trace (VT) of the line is 40 mm above the XY line (in the VP). The horizontal distance between the projectors of HT and VT is 50 mm. The front view of end P is 5 mm above the XY line. The top view of end Q is 20 mm below the XY line. (10)
5. Draw the projections of a circle of 5 cm diameter having its plane vertical and inclined at 30° to the V.P. Its center is 3cm above the H.P. and 2 cm in front of the V.P. (10)



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech, First Year

End Semester Examinations, April-2026

Course Name: **ENGINEERING DRAWING**

Date: 23-04-2026

Duration: 2 Hours

Course Code: **ME101**

Time: 2:00 PM to 4:00 PM

Max. Marks: 100

ANSWER ALL QUESTIONS

1.

(a) State whether the following statements are true or false (5).

- If the front view (F.V.) of an object appears as a square, then a pentagonal true shape may be possible.
- For a pyramid with n corners, the minimum number of non-overlapping corners visible in the F.V is $(n - 2)$ when n is odd, and $(n - 1)$ when n is even, assuming the apex is always visible.
- The true shape of the section of a hexagonal prism will be a rectangle only when the cutting plane is parallel to the axis of the prism.
- The given figure (Fig.1) represents the development of the lateral surface of a truncated cylinder.
- The length of a line inclined to the isometric axis is represented in its true length in an isometric view.

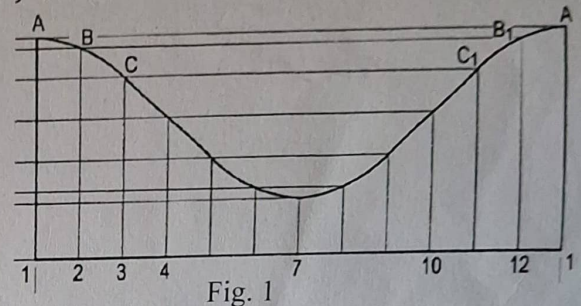
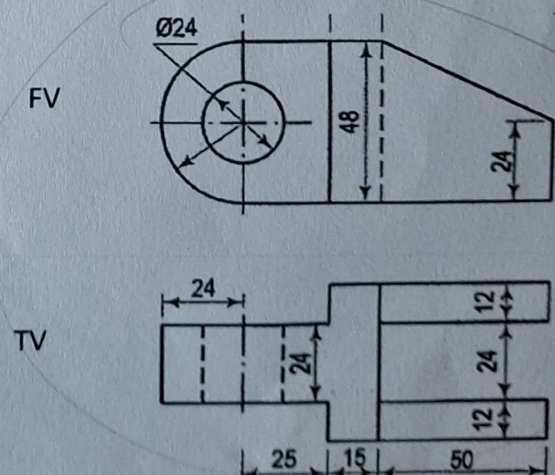


Fig. 1

- The top view of a pentagonal plate is a regular pentagon having edge length of 30 mm with one of the sides inclined at 30° to the reference line. Additionally, the front view is a line inclined at 45° to the reference line. Determine the true shape. (15)
- Draw the projections of a regular pentagonal pyramid with a base side of 30 mm and an axis length of 50 mm. One edge of the base lies on the H.P., and the axis is inclined at 30° to H.P. The apex lies on the V.P., and the axis is inclined at 60° to the V.P. (20)
- A pentagonal prism, 30 mm base side & 60 mm axis is standing on HP on its base with one side of the base perpendicular to VP. It is cut by a section plane inclined at 30° to the HP, through mid-point of axis. Draw FV, sec. TV & sec. Side view. Also draw true shape of section and development of lateral surface of remaining solid. (30)

- The front view and top view of an object are shown below. Draw the isometric view of the object. (30)



***** GOOD LUCK *****



NATIONAL INSTITUTE OF TECHNOLOGY GOA
Cuncolim, Salcete, South Goa-403703
Programme Name: B.Tech.
End Semester Examinations, Dec-2025

Roll No	2	5	C	S	E	1	8	9	2	2
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Course Name: ENGINEERING DRAWING
Date: 08-12-2025
Duration: 2 Hours

Course Code: ME101
Time: 3:00 PM to 5:00 PM
Max. Marks: 100

ANSWER ALL QUESTIONS

- A) What is the difference between an isometric projection and isometric view?
B) When will a section obtained be equivalent to the true shape?
C) What is the maximum number of sides that can be obtained in the cross-section of a regular pentagonal pyramid when it is cut by a single plane? Explain.

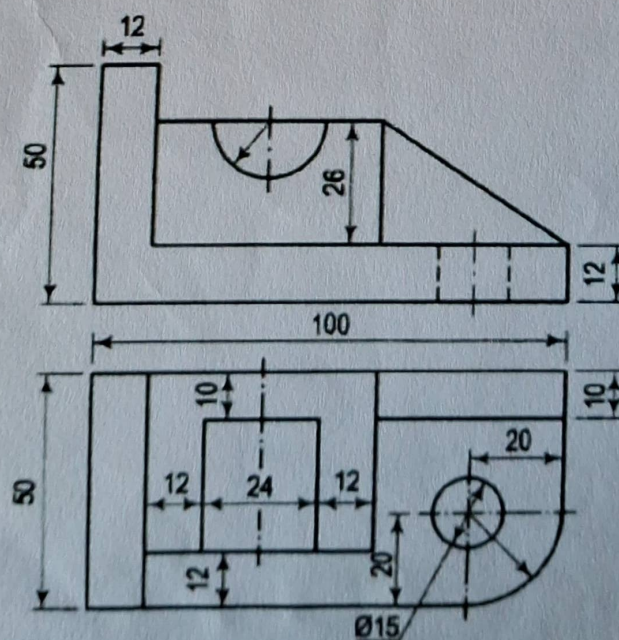
(10)
- A square ABCD of 50 mm side has its corner A in the H.P., its diagonal AC inclined at 30° to the H.P. and the diagonal BD inclined at 45° to the V.P. and parallel to the H.P. Draw its projections.

(10)
- A regular hexagonal pyramid, base 25 mm side and lateral edge (apex to base corners) length of 56 mm, has one of its triangular faces in the V.P. and the edge of the base contained by that face makes an angle of 30° with the H.P. Draw its projections. Find the angles made by the axis with both H.P. and V.P.

(25)
- A pentagonal prism, 30 mm base side & 60 mm axis is standing on HP on its base with one side of the base perpendicular to VP. It is cut by a section plane inclined at 45° to the HP, through mid-point of axis. Draw FV, sec. TV & sec. Side view. Also draw true shape of section and development of lateral surface of remaining solid.

(30)
- Draw the isometric view of the orthographic projection shown below.

(25)



***** GOOD LUCK *****